

Notes on The Lattice Boltzmann Method (LBM)

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NB: Notes for Chapter 1.2, Chapter 1.3, Chapter 3.1 and Chapter 3.2 from the Lattice Boltzmann Method: Principles and Practice textbook. Anything in purple has a footnote with a question. Boxed in blue are summary boxes with key points.

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Definitions A–Z

Average thermal velocity of the molecules	Lattice units
BGK collision operator	Law of Similarity
BGK operator for LBM	Local mean velocity
Boltzmann equation	Mach number
Cauchy momentum equation	Mass density
Collision operator	Maxwell–Boltzmann distribution
Discrete-velocity distribution function	Mean free path
Distribution function	Momentum density
Entropy density	Momentum flux tensor
Equilibrium distribution	Particle populations
Euler energy equation	Particle velocity
Euler momentum equation	Pressure as a moment of the distribution function
Expression for the equilibrium distribution	Relative velocity
Heat flux	Relaxation time
Internal energy density	Reynolds number
Internal energy equation	Stress tensor
Kinematic viscosity	Total energy equation
Knudsen number	Total energy density
Lattice Boltzmann equation	

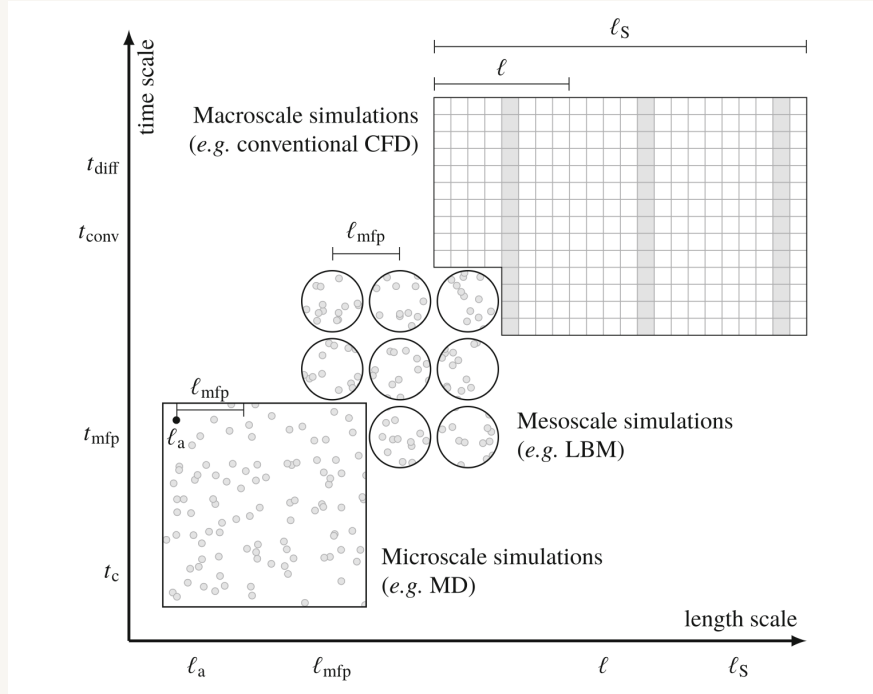
1 Basics of Hydrodynamics and Kinetic Theory

1.1 Relevant Scales

Key Points:

- Outline of the length and time scales in kinetic theory.
- Law of Similarity.

We have the following hierarchy of scales.



Typically, we have the ordering of lengths as,

$$\ell_a \ll \ell_{\text{mfp}} \ll \ell \leq \ell_s$$

where the scales are:

- (i) ℓ_a – the size of an atom or molecule.
- (ii) ℓ_{mfp} – the **mean free path**: the distance travelled between two successive collisions.
- (iii) ℓ – the gradient scale.
- (iv) ℓ_s – the system size.

Similarly, we have the time **scales**¹:

¹What actually are the time scales t_{conv} and t_{diff} ?

- (i) $t_c \sim \frac{\ell_a}{v_T}$ – the collision time.
- (ii) $t_{\text{mfp}} \sim \frac{\ell_{\text{mfp}}}{v_T}$ – the mean flight time between collisions.
- (iii) $t_{\text{conv}} \sim \frac{\ell}{u}$.
- (iv) $t_{\text{diff}} \sim \frac{\ell^2}{\nu}$.
- (v) $t_{\text{sound}} \sim \ell/c_s$ – the acoustic time scale.

where c_s is the speed of sound in the fluid, ν is the [kinematic viscosity](#) which relates to the dynamic shear viscosity by $\eta = \rho\nu$ and $v_T = (k_B T/m)^{1/2}$ is the [average thermal velocity of the molecules](#).

The following numbers are useful:

1. [Reynolds number](#) $\text{Re} = \frac{u\ell}{\nu}$: high and low numbers being of interest.
2. [Mach number](#) $\text{Ma} = \frac{u}{c_s}$: steady flow with $\text{Ma} \leq 0.1$ assumed to be incompressible.
3. [Knudsen number](#) $\text{Kn} = \frac{\ell_{\text{mfp}}}{\ell} = \alpha \frac{\text{Ma}}{\text{Re}}$: for $\text{Kn} \ll 1$ Navier-Stokes is valid and for $\text{Kn} \geq 1$ use kinetic theory.

The [Law of Similarity](#) states that fluids with the same Re , Ma and Kn numbers will have the same physics once rescaled.

However, typically not all dimensionless numbers can be the same when comparing fluids at smaller and larger length or time scales

For instance, nondimensionalising the incompressible Navier-Stokes equation using, in the end it is enough to use the Reynolds number to say two fluids are comparable.

$$\mathbf{u}^* = \frac{\mathbf{u}}{V}, \quad p^* = \frac{p}{\rho V^2}, \quad \mathbf{F}^* = \frac{\mathbf{F}\ell}{\rho V^2}, \quad \frac{\partial}{\partial t^*} = \frac{\ell}{V} \frac{\partial}{\partial t}, \quad \nabla^* = \ell \nabla \quad (1.34)$$

where V is the mean fluid velocity,

$$\begin{aligned} \rho \frac{D\mathbf{u}}{Dt} &= -\nabla p + \eta \Delta \mathbf{u} + \mathbf{F} \\ \iff \rho \left(\frac{V}{\ell} \frac{\partial(V\mathbf{u}^*)}{\partial t^*} + \left(V\mathbf{u}^* \cdot \frac{\nabla^*}{\ell} \right) V\mathbf{u}^* \right) &= -\frac{\nabla^*}{\ell} (\rho V^2 p^*) + \eta \frac{\Delta^*}{\ell^2} (V\mathbf{u}^*) + \frac{\rho V^2}{\ell} \mathbf{F}^* \\ \iff \frac{\rho V^2}{\ell} \left(\frac{\partial \mathbf{u}^*}{\partial t^*} + (\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* \right) &= \frac{\rho V^2}{\ell} \left(-\nabla^* p^* + \frac{\eta}{\rho \ell V} \Delta^* \mathbf{u}^* + \mathbf{F}^* \right) \\ \iff \frac{\partial \mathbf{u}^*}{\partial t^*} + (\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* &= -\nabla^* p^* + \frac{\nu}{\ell V} \Delta^* \mathbf{u}^* + \mathbf{F}^* \\ \iff \frac{D\mathbf{u}^*}{Dt^*} &= -\nabla^* p^* + \frac{1}{\text{Re}} \Delta^* \mathbf{u}^* + \mathbf{F}^* \end{aligned} \quad (1.35)$$

1.2 Kinetic Theory

1.2.1 Introduction

The following description is best for dilute gases and less suitable for dense gases/liquids. This is due to the theory being a mesoscopic description, focusing on the distribution of particles, which requires $t_c \ll t_{\text{mfp}}$ - that is, the particles spend very little time actually colliding.

1.2.2 The Distribution Function and its Moments

Key Points:

- $f(\mathbf{x}, \boldsymbol{\xi}, t)$ is the density of particles and its moments correspond to physical properties of the fluid.

The **distribution function** $f(\mathbf{x}, \boldsymbol{\xi}, t)$ is the generalisation of the density ρ which also takes into account the microscopic particle velocity $\boldsymbol{\xi} = (\xi_x, \xi_y, \xi_z)$; this represents the density of particles with velocity $\boldsymbol{\xi}$ at position $\mathbf{x}$ at time t .

Example 1.1 Consider a gas in a box of size $V = L_x \times L_y \times L_z$. We can find expressions using the density,

$$\text{Total mass} = \int_V \rho d^3\mathbf{x}$$

$$\text{Total mass in the left side of the box} = \int_{x=0}^{x=L_x/2} \rho d^3\mathbf{x}$$

but using the distribution function we can gain even more specific information. For instance, the total mass of right moving particles in the left hand side of the box,

$$\int_{x=0}^{x=L_x/2} \int_{\xi_x > 0} f(\mathbf{x}, \boldsymbol{\xi}, t) d^3\boldsymbol{\xi} d^3\mathbf{x} \quad (1.37)$$

We can further recover macroscopic properties using the distribution function,

$$\text{Mass Density} = \rho(\mathbf{x}, t) = \int f(\mathbf{x}, \boldsymbol{\xi}, t) d^3\boldsymbol{\xi} \quad (1.38a)$$

$$\text{Momentum Density} = \rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t) = \int \boldsymbol{\xi} f(\mathbf{x}, \boldsymbol{\xi}, t) d^3\boldsymbol{\xi} \quad (1.38b)$$

$$\text{Total Energy Density} = \rho(\mathbf{x}, t)E(\mathbf{x}, t) = \frac{1}{2} \int |\boldsymbol{\xi}|^2 f(\mathbf{x}, \boldsymbol{\xi}, t) d^3\boldsymbol{\xi} \quad (1.38c)$$

$$\text{Internal Energy Density} = \rho(\mathbf{x}, t)e(\mathbf{x}, t) = \frac{1}{2} \int |\mathbf{v}|^2 f(\mathbf{x}, \boldsymbol{\xi}, t) d^3\boldsymbol{\xi} \quad (1.38d)$$

where we have defined the **relative velocity**,

$$\mathbf{v}(\mathbf{x}, t) := \boldsymbol{\xi}(\mathbf{x}, t) - \mathbf{u}(\mathbf{x}, t) \quad (1.39)$$

where $\boldsymbol{\xi}$ is the [particle velocity](#) and $\mathbf{u}$ is the [local mean velocity](#).

Exercise 1.5 Consider a somewhat unrealistic spatially homogeneous gas where all particles are moving with the same velocity $\mathbf{u}$ so that the distribution function is $f(\mathbf{x}, \boldsymbol{\xi}, t) = \rho \delta(\boldsymbol{\xi} - \mathbf{u})$.

Verify from its moments that its density is ρ and its momentum density is $\rho \mathbf{u}$. Additionally, find its moments of total energy density and internal energy density.

Solution Calculating the required integrals,

$$\begin{aligned} \int_{\mathbb{R}^3} f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} &= \int_{\mathbb{R}^3} \rho \delta(\boldsymbol{\xi} - \mathbf{u}) d^3 \boldsymbol{\xi} = \rho \int_{\mathbb{R}^3} \delta(\boldsymbol{\xi} - \mathbf{u}) d^3 \boldsymbol{\xi} = \rho \\ \int_{\mathbb{R}^3} \boldsymbol{\xi} f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} &= \int_{\mathbb{R}^3} \boldsymbol{\xi} \rho \delta(\boldsymbol{\xi} - \mathbf{u}) d^3 \boldsymbol{\xi} = \rho \int_{\mathbb{R}^3} \boldsymbol{\xi} \delta(\boldsymbol{\xi} - \mathbf{u}) d^3 \boldsymbol{\xi} = \rho \mathbf{u} \end{aligned}$$

using the appropriate properties of the three dimensional delta function.

Moreover, we can find the total energy density,

$$\frac{1}{2} \int_{\mathbb{R}^3} |\boldsymbol{\xi}|^2 f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} = \frac{1}{2} \int_{\mathbb{R}^3} |\boldsymbol{\xi}|^2 \rho \delta(\boldsymbol{\xi} - \mathbf{u}) d^3 \boldsymbol{\xi} = \frac{1}{2} \rho \int_{\mathbb{R}^3} |\boldsymbol{\xi}|^2 \delta(\boldsymbol{\xi} - \mathbf{u}) d^3 \boldsymbol{\xi} = \frac{1}{2} \rho |\mathbf{u}|^2$$

and the internal energy density,

$$\frac{1}{2} \int_{\mathbb{R}^3} |\mathbf{v}|^2 f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} = \frac{1}{2} \int_{\mathbb{R}^3} |\boldsymbol{\xi} - \mathbf{u}|^2 \rho \delta(\boldsymbol{\xi} - \mathbf{u}) d^3 \boldsymbol{\xi} = \frac{1}{2} \rho |\mathbf{u} - \mathbf{u}|^2 = 0$$

Exercise 1.6

(a) Show that the relative velocity moment of f is

$$\int \mathbf{v} f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} = \mathbf{0}. \quad (1.40a)$$

(b) Using this and the identity $|\mathbf{v}|^2 = |\boldsymbol{\xi}|^2 - 2(\boldsymbol{\xi} \cdot \mathbf{u}) + |\mathbf{u}|^2$, show that the total and internal energy densities at a given position and time are related as

$$\rho e = \rho E - \frac{1}{2} \rho |\mathbf{u}|^2. \quad (1.40b)$$

Solution (a) By direct integration and linearity of the integral,

$$\int \mathbf{v} f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} = \int \boldsymbol{\xi} f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} - \mathbf{u} \int f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} = \rho \mathbf{u} - \rho \mathbf{u} = \mathbf{0}.$$

(b) Using the definition of the internal energy density,

$$\begin{aligned}
 \rho e &= \frac{1}{2} \int |\mathbf{v}|^2 f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} = \frac{1}{2} \int (|\boldsymbol{\xi}|^2 - 2(\boldsymbol{\xi} \cdot \mathbf{u}) + |\mathbf{u}|^2) f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} \\
 &= \frac{1}{2} \int |\boldsymbol{\xi}|^2 f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} - \int (\boldsymbol{\xi} \cdot \mathbf{u}) f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} + \frac{1}{2} |\mathbf{u}|^2 \int f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} \\
 &= \rho E - \mathbf{u} \cdot \int \boldsymbol{\xi} f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} + \frac{1}{2} \rho |\mathbf{u}|^2 \\
 &= \rho E - \rho(\mathbf{u} \cdot \mathbf{u}) + \frac{1}{2} \rho |\mathbf{u}|^2 = \rho E - \rho |\mathbf{u}|^2 + \frac{1}{2} \rho |\mathbf{u}|^2 \\
 &= \rho E - \frac{1}{2} \rho |\mathbf{u}|^2.
 \end{aligned}$$

If you are unconvinced for the third line, think about the dot product in terms of components and the fact that $\mathbf{u}$ is independent of $\boldsymbol{\xi}$ to bring each component out of the integral. Once there, reintroduce the dot product.

Doing a suspiciously similar hand wavy approach as in A-Level physics leads to,

$$\rho e = \frac{3}{2} \rho R T = \frac{3}{2} p. \quad (1.41)$$

This results in an expression for the [pressure as a moment of the distribution function](#),

$$p = \rho R T = \frac{2}{3} \rho e = \frac{1}{3} \int |\mathbf{v}|^2 f(\mathbf{x}, \boldsymbol{\xi}, t) d^3 \boldsymbol{\xi} \quad (1.42)$$

1.2.3 The Equilibrium Distribution Function

Key Points:

- $f(\mathbf{x}, \boldsymbol{\xi}, t) \rightarrow f^{\text{eq}}(\mathbf{x}, \boldsymbol{\xi}, t) = f^{\text{eq}}(\mathbf{x}, |\mathbf{v}|, t)$.
- The equilibrium function has the form (among others)

$$f^{\text{eq}}(\mathbf{x}, |\mathbf{v}|, t) = \rho \left(\frac{1}{2\pi R T} \right)^{3/2} e^{-|\mathbf{v}|^2 / (2 R T)}.$$

We will assume that collisions tend to even out the angular distribution of particles velocities in a gas around the mean velocity $\mathbf{u}$.

When a gas has been left alone for a sufficient amount of time, we may assume that the distribution function $f(\mathbf{x}, \boldsymbol{\xi}, t)$ will reach an [equilibrium distribution](#) $f^{\text{eq}}(\mathbf{x}, \boldsymbol{\xi}, t)$.

The equilibrium distribution is isotropic (the same in all directions) in velocity space around $\boldsymbol{\xi} = \mathbf{u}$. That is in a reference frame moving with velocity $\mathbf{u}$, the equilibrium distribution can be expressed as $f^{\text{eq}}(\mathbf{x}, |\mathbf{v}|, t)$.

Moreover, we will reduce the equilibrium distribution into a separable form

$$f^{\text{eq}}(|\mathbf{v}|^2) = f_{1D}^{\text{eq}}(v_x^2) f_{1D}^{\text{eq}}(v_y^2) f_{1D}^{\text{eq}}(v_z^2) \quad (1.44a)$$

that is the 3D equilibrium distribution is the product of three 1D equilibrium distributions.

If we hold the magnitude $|\mathbf{v}|^2 = v_x^2 + v_y^2 + v_z^2$ constant,

$$f^{\text{eq}}(|\mathbf{v}|^2) = \text{const} \Rightarrow \ln f^{\text{eq}}(v_x^2) + \ln f^{\text{eq}}(v_y^2) + \ln f^{\text{eq}}(v_z^2) = \text{const}. \quad (1.44b)$$

which is true if $\ln f_{1D}^{\text{eq}}(v_i^2) = a + bv_i^2$, with a and b being constants and $i \in \{x, y, z\}$. Hence,

$$\ln f^{\text{eq}}(v_x^2) + \ln f^{\text{eq}}(v_y^2) + \ln f^{\text{eq}}(v_z^2) = 3a + b(v_x^2 + v_y^2 + v_z^2) = \text{const}, \quad (1.44c)$$

and the full 3D equilibrium distribution is of the form

$$f^{\text{eq}}(|\mathbf{v}|) = e^{3a} e^{b|\mathbf{v}|^2}. \quad (1.44d)$$

To find a and b , we demand that f^{eq} has the same moments of density and energy as f .

Thus, we have an [expression for the equilibrium distribution](#) as

$$\begin{aligned} f^{\text{eq}}(\mathbf{x}, |\mathbf{v}|, t) &= \rho \left(\frac{3}{4\pi e} \right)^{3/2} e^{-3|\mathbf{v}|^2/(4e)} = \rho \left(\frac{\rho}{2\pi\rho} \right)^{3/2} e^{-p|\mathbf{v}|^2/(2\rho)} \\ &= \rho \left(\frac{1}{2\pi RT} \right)^{3/2} e^{-|\mathbf{v}|^2/(2RT)}. \end{aligned} \quad (1.45)$$

which is also known as the [Maxwell-Boltzmann distribution](#)

1.2.4 The Boltzmann Equation and the Collision Operator

Key Points:

- The Boltzmann Equation

$$\frac{\partial f}{\partial t} + \xi_\beta \frac{\partial f}{\partial x_\beta} + \frac{F_\beta}{\rho} \frac{\partial f}{\partial \xi_\beta} = \Omega(f).$$

- $\Omega(f) = \frac{df}{dt}$ is the collision operator but we can have other operators.

Finding the total derivative of $f = f(\mathbf{x}, \boldsymbol{\xi}, t)$,

$$\frac{df}{dt} = \frac{\partial f}{\partial t} \frac{dt}{dt} + \frac{\partial f}{\partial x_\beta} \frac{dx_\beta}{dt} + \frac{\partial f}{\partial \xi_\beta} \frac{d\xi_\beta}{dt} \quad (1.46)$$

which upon inspection yields,

$$\frac{\partial f}{\partial t} + \xi_\beta \frac{\partial f}{\partial x_\beta} + \frac{F_\beta}{\rho} \frac{\partial f}{\partial \xi_\beta} = \Omega(f) \quad (1.47)$$

where $\Omega(f) = \frac{df}{dt}$ and we have used $\frac{dt}{dt} = 1$, $\frac{dx_\beta}{dt} = \xi_\beta$ and $\frac{\partial \xi_\beta}{\partial t} = \frac{F_\beta}{\rho}$ the final being Newton's second law which is known as the [Boltzmann equation](#). Moreover, the source term $\Omega(f)$ is called the [collision operator](#).

Since collisions conserve mass, momentum and translational energy, we can represent these conservations in the form of moments of the collision operator,

$$\text{Mass Conservation : } \int \Omega(f) d^3\xi = 0 \quad (1.48a)$$

$$\text{Momentum Conservation : } \int \xi \Omega(f) d^3\xi = \mathbf{0} \quad (1.48b)$$

$$\text{Total Energy Conservation: } \int |\xi|^2 \Omega(f) d^3\xi = 0 \quad (1.48c)$$

$$\text{Internal Energy Conservation : } \int |\mathbf{v}|^2 \Omega(f) d^3\xi = 0 \quad (1.48d)$$

The Bhatnagar–Gross–Krook (BGK) [operator](#) is given by:

$$\Omega(f) = -\frac{1}{\tau}(f - f^{\text{eq}}) \quad (1.49)$$

with time constant τ being the [relaxation time](#), determining the speed of this equilibration.

Exercise 1.10 For a force-free, spacially homogeneous case $f(\mathbf{x}, \xi, t) \rightarrow f(\xi, t)$, show that the BGK operator relaxes an initial distribution function $f(\xi, t = 0)$ exponentially to the equilibrium distribution $f^{\text{eq}}(\xi)$ as

$$f(\xi, t) = f^{\text{eq}}(\xi) + [f(\xi, 0) - f^{\text{eq}}(\xi)]e^{-t/\tau}. \quad (1.50)$$

Solution Starting with the Boltzmann equation for the BGK operator,

$$\frac{\partial f}{\partial t} + \xi_\beta \frac{\partial f}{\partial x_\beta} + \frac{F_\beta}{\rho} \frac{\partial f}{\partial \xi_\beta} = -\frac{1}{\tau}(f - f^{\text{eq}})$$

under the assumptions our equation reduces to,

$$-\frac{1}{\tau}(f - f^{\text{eq}}) = \frac{\partial f}{\partial t} \iff \frac{\partial f}{\partial t} + \frac{1}{\tau}f = \frac{1}{\tau}f^{\text{eq}}$$

which can be solved with an integrating factor $I = e^{t/\tau}$,

$$\frac{\partial f}{\partial t} + \frac{1}{\tau}f = \frac{1}{\tau}f^{\text{eq}} \iff \frac{\partial}{\partial t}(If) = \frac{1}{\tau}f^{\text{eq}}I$$

Using that $f^{\text{eq}} = f^{\text{eq}}(\xi)$,

$$If = \frac{1}{\tau}f^{\text{eq}} \int e^{t/\tau} dt = f^{\text{eq}}e^{t/\tau} + c \Rightarrow f(\xi, t) = f^{\text{eq}} + ce^{-t/\tau}$$

and we can find the constant c in terms of $f(\xi, 0)$,

$$\Rightarrow f(\xi, t) = f^{\text{eq}}(\xi) + [f(\xi, 0) - f^{\text{eq}}(\xi)]e^{-t/\tau}.$$

1.2.5 Macroscopic Conservation Equations

Key Points:

- Mass conservation

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_\beta)}{\partial x_\beta} = 0.$$

- Momentum conservation

$$\frac{\partial(\rho u_\alpha)}{\partial t} + \frac{\partial \Pi_{\alpha\beta}}{\partial x_\beta} = F_\alpha.$$

- Cauchy momentum equation

$$\frac{\partial(\rho u_\alpha)}{\partial t} + \frac{\partial(\rho u_\alpha u_\beta)}{\partial x_\beta} = \frac{\partial \sigma_{\alpha\beta}}{\partial x_\beta} + F_\alpha.$$

- Total energy equation

$$\frac{\partial(\rho E)}{\partial t} + \frac{\partial(\rho u_\beta E)}{\partial x_\beta} = \frac{\partial(u_\alpha \sigma_{\alpha\beta})}{\partial x_\beta} + F_\beta u_\beta - \frac{\partial q_\beta}{\partial x_\beta}.$$

- Internal energy equation

$$\frac{\partial(\rho e)}{\partial t} + \frac{\partial(\rho u_\beta e)}{\partial x_\beta} = \sigma_{\alpha\beta} \frac{\partial u_\beta}{\partial x_\beta} - \frac{\partial q_\beta}{\partial x_\beta}.$$

We define the notation:

$$\begin{aligned} \Pi_0 &= \int f \, d^3 \boldsymbol{\xi} = \rho, & \Pi_\alpha &= \int \xi_\alpha f \, d^3 \boldsymbol{\xi} = \rho u_\alpha, \\ \Pi_{\alpha\beta} &= \int \xi_\alpha \xi_\beta f \, d^3 \boldsymbol{\xi}, & \Pi_{\alpha\beta\gamma} &= \int \xi_\alpha \xi_\beta \xi_\gamma f \, d^3 \boldsymbol{\xi}. \end{aligned} \quad (1.51)$$

Using integration by parts, we further have:

$$\begin{aligned} \int \frac{\partial f}{\partial \xi_\beta} \, d^3 \boldsymbol{\xi} &= 0, \\ \int \xi_\alpha \frac{\partial f}{\partial \xi_\beta} \, d^3 \boldsymbol{\xi} &= - \int \frac{\partial \xi_\alpha}{\partial \xi_\beta} f \, d^3 \boldsymbol{\xi} = -\rho \delta_{\alpha\beta}, \\ \int \xi_\alpha \xi_\alpha \frac{\partial f}{\partial \xi_\beta} \, d^3 \boldsymbol{\xi} &= - \int \frac{\partial(\xi_\alpha \xi_\alpha)}{\partial \xi_\beta} f \, d^3 \boldsymbol{\xi} = -2\rho u_\beta, \end{aligned} \quad (1.52)$$

Mass Conservation Equation

Using the Boltzmann equation, we can find the continuity equation by directly integrating the equation over velocity space,

$$\begin{aligned} \int \left(\frac{\partial f}{\partial t} + \xi_\beta \frac{\partial f}{\partial x_\beta} + \frac{F_\beta}{\rho} \frac{\partial f}{\partial \xi_\beta} \right) d^3 \boldsymbol{\xi} &= \int \Omega(f) \, d^3 \boldsymbol{\xi} \\ \Rightarrow \frac{\partial}{\partial t} \int f \, d^3 \boldsymbol{\xi} + \frac{\partial}{\partial x_\beta} \int \xi_\beta f \, d^3 \boldsymbol{\xi} + \frac{F_\beta}{\rho} \int \frac{\partial f}{\partial \xi_\beta} \, d^3 \boldsymbol{\xi} &= \int \Omega(f) \, d^3 \boldsymbol{\xi}. \end{aligned} \quad (1.53)$$

which results in the continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_\beta)}{\partial x_\beta} = 0 \quad (1.54)$$

Momentum Conservation Equation Using the Boltzmann equation, we can find its first moment

$$\int \left(\xi_\alpha \frac{\partial f}{\partial t} + \xi_\alpha \xi_\beta \frac{\partial f}{\partial x_\beta} + \xi_\alpha \frac{F_\beta}{\rho} \frac{\partial f}{\partial \xi_\beta} \right) d^3 \xi = \int \xi_\alpha \Omega(f) d^3 \xi$$

to find the momentum conservation equation,

$$\frac{\partial(\rho u_\alpha)}{\partial t} + \frac{\partial \Pi_{\alpha\beta}}{\partial x_\beta} = F_\alpha \quad (1.55)$$

where $\Pi_{\alpha\beta}$ is the [momentum flux tensor](#) defined in (1.51).

Using the decomposition of the momentum flux tensor,

$$\Pi_{\alpha\beta} = \rho u_\alpha u_\beta + \int v_\alpha v_\beta f d^3 \xi \quad (1.56)$$

we can calculate the second moment of the Boltzmann equation

$$\int \left(\xi_\alpha^2 \frac{\partial f}{\partial t} + \xi_\alpha^2 \xi_\beta \frac{\partial f}{\partial x_\beta} + \xi_\alpha^2 \frac{F_\beta}{\rho} \frac{\partial f}{\partial \xi_\beta} \right) d^3 \xi = \int \xi_\alpha^2 \Omega(f) d^3 \xi$$

to find,

$$\frac{\partial(\rho u_\alpha)}{\partial t} + \frac{\partial(\rho u_\alpha u_\beta)}{\partial x_\beta} = \frac{\partial \sigma_{\alpha\beta}}{\partial x_\beta} + F_\alpha \quad (1.57)$$

which is known as the [Cauchy momentum equation](#) and $\sigma_{\alpha\beta}$ is the [stress tensor](#)

$$\sigma_{\alpha\beta} = - \int v_\alpha v_\beta f d^3 \xi$$

Energy Conservation Equation Finally, finding the trace of the second moment of the Boltzmann equation

$$\int \left(\xi_\alpha \xi_\alpha \frac{\partial f}{\partial t} + \xi_\alpha \xi_\alpha \xi_\beta \frac{\partial f}{\partial x_\beta} + \xi_\alpha \xi_\alpha \frac{F_\beta}{\rho} \frac{\partial f}{\partial \xi_\beta} \right) d^3 \xi = \int \xi_\alpha \xi_\alpha \Omega(f) d^3 \xi$$

leaves us with,

$$\frac{\partial(\rho E)}{\partial t} + \frac{1}{2} \frac{\partial}{\partial \Pi_{\alpha\beta}} x_\beta = F_\beta u_\beta \quad (1.59)$$

We can simplify this in two steps. Firstly, by splitting the moment just as before which results in

$$\frac{\partial(\rho E)}{\partial t} + \frac{\partial(\rho u_\beta E)}{\partial x_\beta} = \frac{\partial(u_\alpha \sigma_{\alpha\beta})}{\partial x_\beta} + F_\beta u_\beta - \frac{\partial q_\beta}{\partial x_\beta} \quad (1.60)$$

which is known as the [total energy equation](#) with $\mathbf{q}$ the [heat flux](#) as the moment

$$q_\beta = \frac{1}{2} \int v_\alpha v_\alpha v_\beta f \, d^3\xi \quad (1.61)$$

Secondly, by eliminating the component $\frac{1}{2}\rho|\mathbf{u}|^2$ from the equation by subtracting (1.57) multiplied by u_α , resulting in the [internal energy equation](#)

$$\frac{\partial(\rho e)}{\partial t} + \frac{\partial(\rho u_\beta e)}{\partial x_\beta} = \sigma_{\alpha\beta} \frac{\partial u_\beta}{\partial x_\alpha} - \frac{\partial q_\beta}{\partial x_\beta}. \quad (1.62)$$

Discussion The mass equation is independent of f whilst the momentum/energy equations do depend on f .

Assuming $f \simeq f^{\text{eq}}$ in (1.58) and (1.61) gives the [Euler momentum equation](#) and the [Euler energy equation](#)

$$\frac{\partial(\rho u_\alpha)}{\partial t} + \frac{\partial(\rho u_\alpha u_\beta)}{\partial x_\beta} = -\frac{\partial p}{\partial x_\alpha} - F_\alpha, \quad \frac{\partial(\rho e)}{\partial t} + \frac{\partial(\rho u_\beta e)}{\partial x_\beta} = -p \frac{\partial u_\beta}{\partial x_\beta}. \quad (1.63)$$

Skipping some analysis, the Boltzmann equation describes the macroscopic behaviour of a fluid. To a zero order approximation ($f \approx f^{\text{eq}}$) giving the Euler model. To a first order approximation ($f \approx f^{\text{eq}} + \varepsilon f^{(1)}$) giving the Navier-Stokes-Fourier model.

1.2.6 Boltzmann's $\mathcal{H}$ -Theorem

Key Points:

- The quantity

$$\mathcal{H} = \int f \ln f \, d^3\xi$$

is decreasing and reaches a minimum at $f = f^{\text{eq}}$.

The [entropy density](#) is denoted ρs .

The quantity

$$\mathcal{H} = \int f \ln f \, d^3\xi \quad (1.67)$$

can only decrease and reaches its minimum value when f reaches its equilibrium f^{eq} . This can be seen by using (1.46) and multiplying with $(1 + \ln f)$, using the reverse chain rule and taking the zeroth moment,

$$\frac{\partial}{\partial t} \int f \ln f \, d^3\xi + \frac{\partial}{\partial x_\alpha} \int \xi_\alpha f \ln f \, d^3\xi = \int \Omega(f) \ln f \, d^3\xi \quad (1.68)$$

and further analysis. This equation also corresponds to,

$$\frac{\partial \mathcal{H}}{\partial t} + \frac{\partial \mathcal{H}_\alpha}{\partial x_\alpha} \leq 0 \quad (1.70)$$

which shows $\mathcal{H}$ is not a conserved quantity and decreases until it reaches equilibrium - this is Boltzmann's $\mathcal{H}$ -theorem. In fact for ideal gases,

$$\rho s = -R\mathcal{H}. \quad (1.71)$$

2 The Lattice Boltzmann Equation

2.1 The Lattice Boltzmann Equation in a Nutshell

Key Points:

- $f \rightarrow f_i(\mathbf{x}, t)$
- Can use moments of f_i to find ρ and $\rho \mathbf{u}$.
- Lattice Boltzmann equation

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) + \Omega_i(\mathbf{x}, t).$$

- BGK operator

$$\Omega_i(f_i) = -\frac{f_i - f_i^{\text{eq}}}{\tau} \Delta t.$$

- The equilibrium distribution

$$f_i^{\text{eq}}(\mathbf{x}, t) = w_i \rho \left(1 + \frac{\mathbf{u} \cdot \mathbf{c}_i}{c_s^2} + \frac{(\mathbf{u} \cdot \mathbf{c}_i)^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right).$$

- Do it like so:
 1. Calculate ρ and $\mathbf{u}$ from the moments of f_i .
 2. Find the equilibrium distribution f_i^{eq} .
 3. Find the post collision distribution f_i^* .
 4. Stream the post collision distribution to update f_i^* .

Overview The basic quantity is the [discrete-velocity distribution function](#) $f_i(\mathbf{x}, t)$, often referred to as the [particle populations](#); this represents the density of particles with velocity $\mathbf{c}_i = (c_{ix}, c_{iy}, c_{iz})$ at position $\mathbf{x}$ and time t .

Similarly, the mass density ρ and momentum density $\rho \mathbf{u}$ at $(\mathbf{x}, t)$ can be found through the moments of f_i :

$$\rho(\mathbf{x}, t) = \sum_i f_i(\mathbf{x}, t), \quad \rho(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) = \sum_i \mathbf{c}_i f_i(\mathbf{x}, t). \quad (3.1)$$

The difference between f_i and the continuous f is that the arguments of f_i are discrete:

- The $\mathbf{c}_i$'s are one of a discrete set of possible velocities $\{\mathbf{c}_i\}$.
- The points $\mathbf{x}$ are positioned as a square lattice in space with spacing Δx .
- f_i is defined at time steps t with time spacing Δt .

The units of Δx and Δt can be any physical units or **lattice units** where $\Delta t = \Delta x = 1$ and we can convert from these units into the appropriate scale.

The velocities $\mathbf{c}_i$ together with weighting coefficients w_i form a velocity set $\{\mathbf{c}_i, w_i\}$ which is denoted DdQq with d the spacial dimension and q the is the set's number of velocities.

In each velocity set, we can find a constant c_s which determines the relation $p = c_s^2 \rho$ and this constant represents the model's speed of sound. In the sets mentioned, $c_s^2 = \frac{1}{3} \frac{\Delta x^2}{\Delta t^2}$.

Discretising the Boltzmann equation in velocity space results in

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) + \Omega_i(\mathbf{x}, t) \quad (3.2)$$

known as the **lattice Boltzmann equation**:

- Particles f_i move with velocity $\mathbf{c}_i$ to a neighbouring point $\mathbf{x} + \mathbf{c}_i \Delta t$ at the next time step $t + \Delta t$.
- Particles are affected by the collision operator Ω_i , which redistributes particles among the populations f_i at each site.

The simplest operator is the **BGK operator**:

$$\Omega_i(f_i) = -\frac{f_i - f_i^{\text{eq}}}{\tau} \Delta t \quad (3.3)$$

which relaxes the population towards an equilibrium rate f_i^{eq} determined by the relaxation time τ , which is given

$$f_i^{\text{eq}}(\mathbf{x}, t) = w_i \rho \left(1 + \frac{\mathbf{u} \cdot \mathbf{c}_i}{c_s^2} + \frac{(\mathbf{u} \cdot \mathbf{c}_i)^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right) \quad (3.4)$$

This is such that it has the same moments as f_i .

The link between LBE and NSE is that the LBE results in macroscopic behaviour according to the NSE with shear viscosity

$$\nu = c_s^2 \left(\tau - \frac{\Delta t}{2} \right) \quad (3.5)$$

The Time Step: Collision and Streaming The full LBGK reads,

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)}{\tau} \Delta t. \quad (3.7)$$

We have two parts:

1. The first part is collision (relaxation),

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) - \frac{\Delta t}{\tau}(f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)) \quad (3.8)$$

where f_i^* represents the distribution function after the collision, often implemented as

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) \left(1 - \frac{\Delta t}{\tau}\right) + f_i^{\text{eq}}(\mathbf{x}, t) \frac{\Delta t}{\tau}. \quad (3.9)$$

2. The second part is streaming (propagation),

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i^*(\mathbf{x}, t)$$